DSA 0604

Data Handling and Visualization

Harish Kumar JN

192524502

Set 21

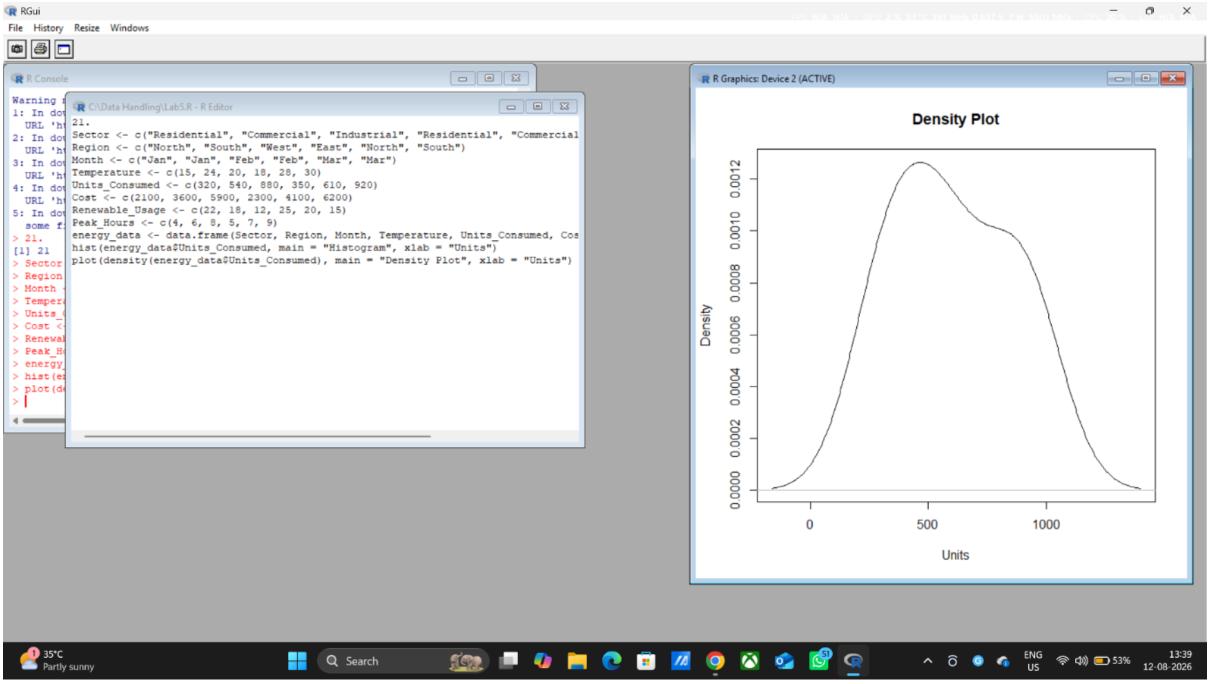
Dataset: Energy_Consumption_Data.csv

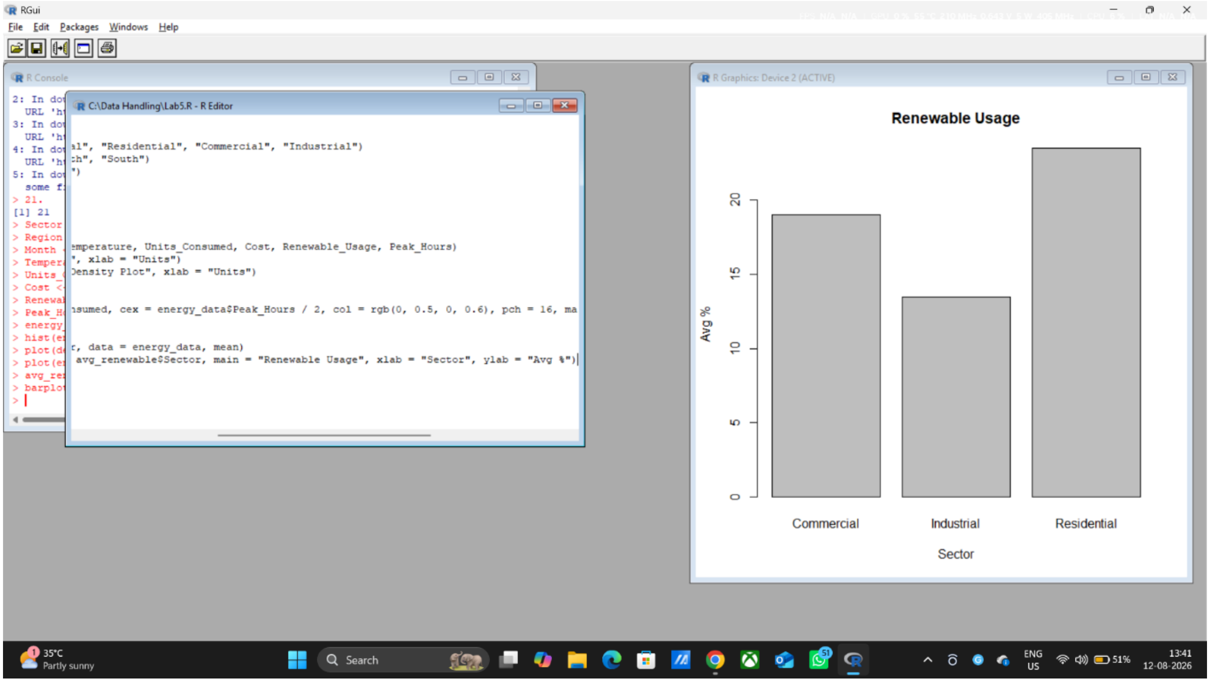
1. Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.

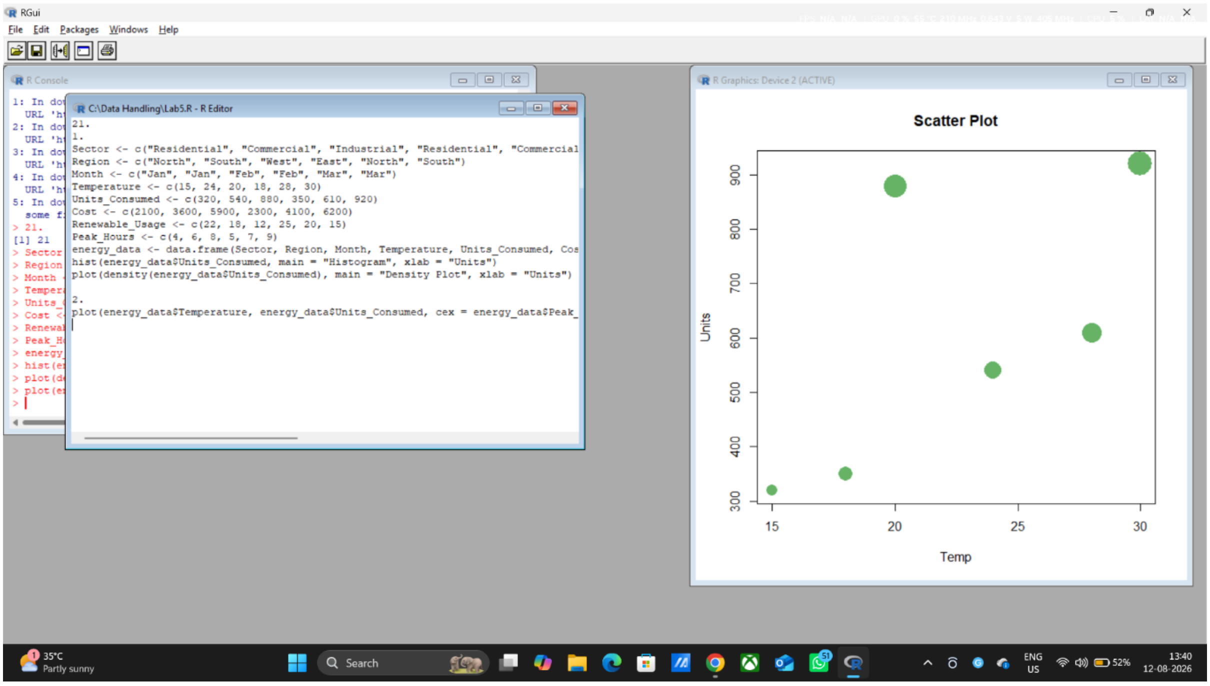
2. Using R programming, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.

3. Using R, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.

4. Import Energy_Consumption_Data.csv into Tableau. Create a line chart for monthly energy usage trends. Create a heatmap (Sector vs Region vs Units). Add filters for Month and Sector. Design an interactive dashboard.







Set 22: Time Series Analysis

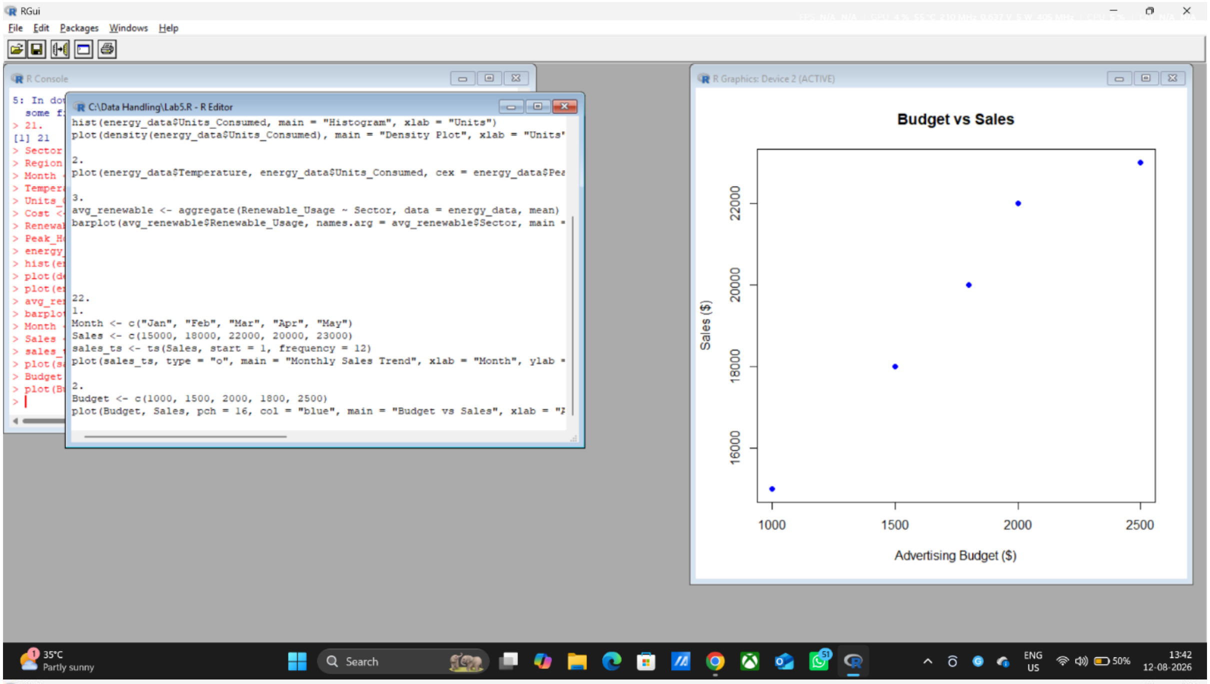
Dataset: Monthly Sales Data

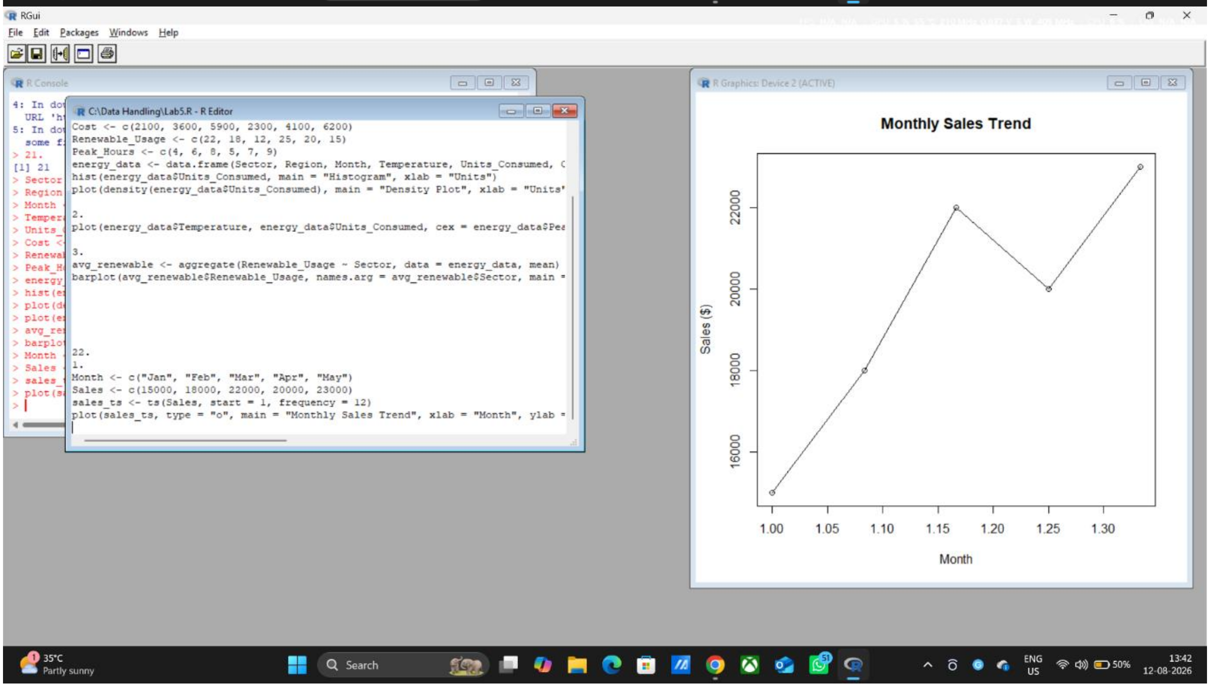
1. In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.

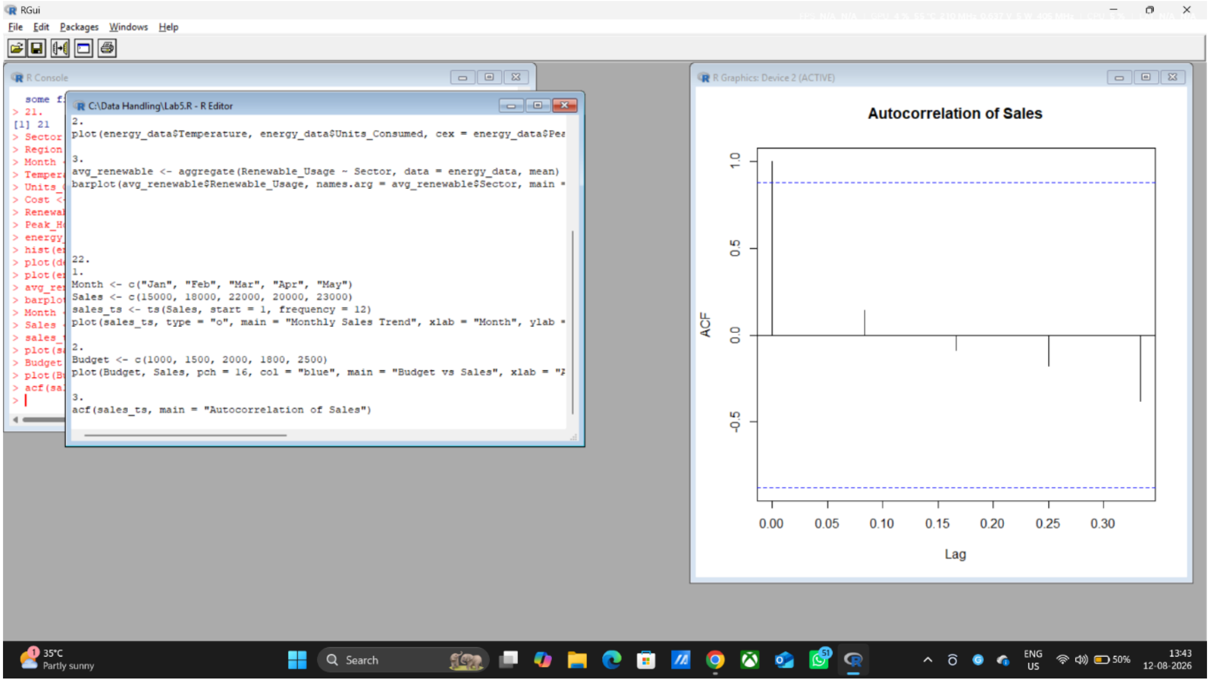
2. In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.

3. In Tableau, build a dashboard combining a line chart showing sales trend and a pie chart displaying the distribution of sales across products.

4. In R, create an autocorrelation plot to identify seasonality in the time series data.







Set 23: Employee Performance Analysis

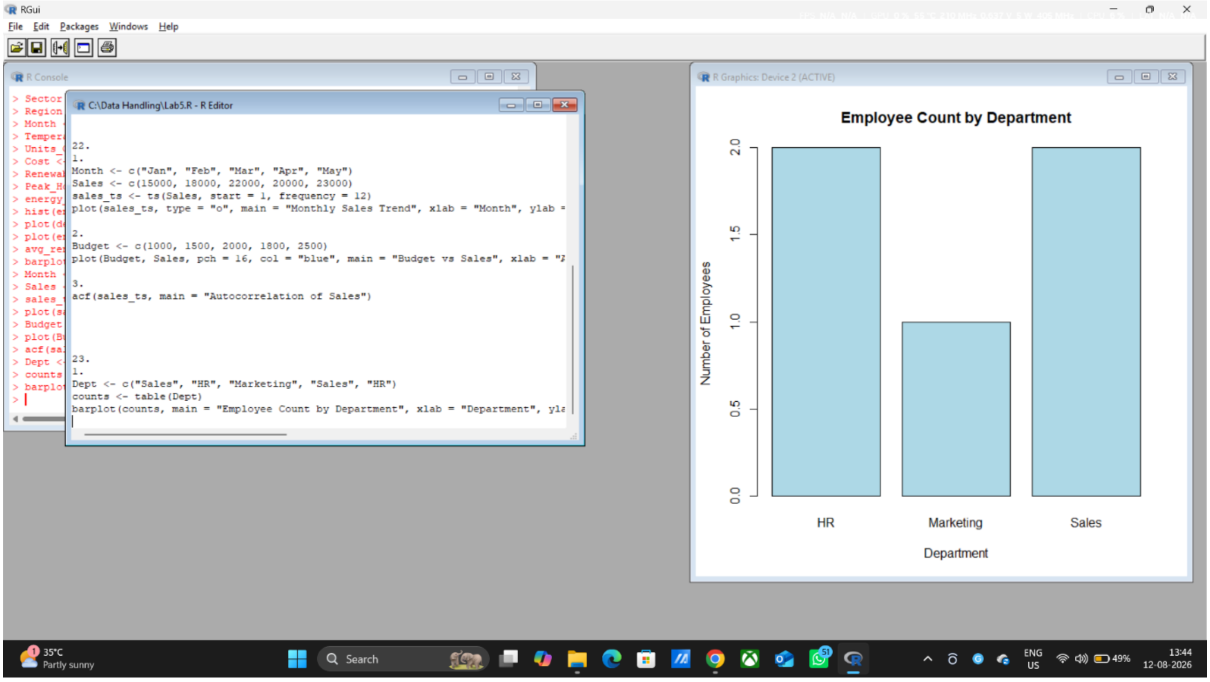
Dataset: Employee Performance

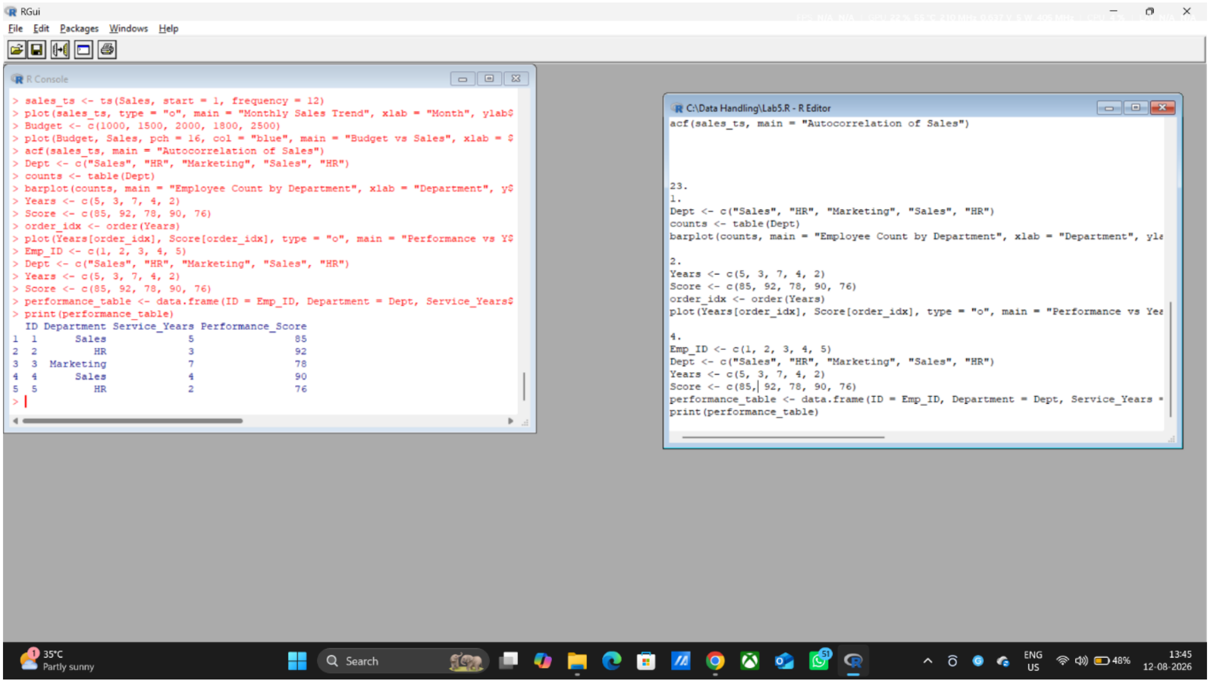
1. In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.

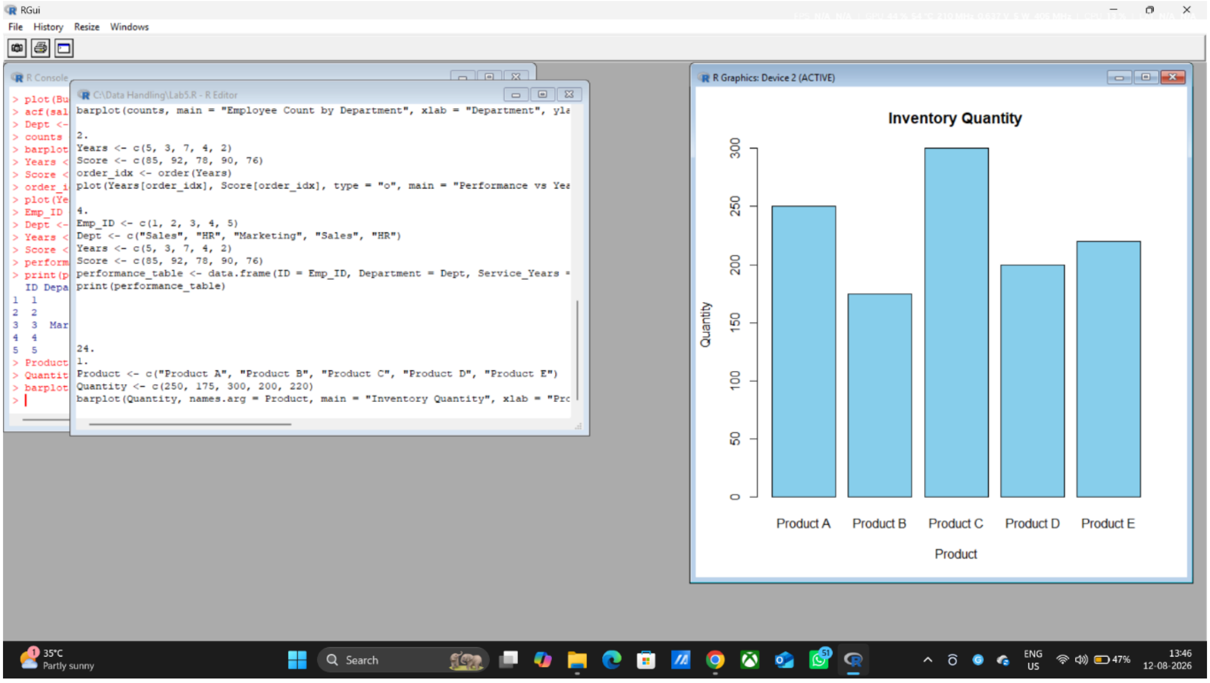
2. In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

3. In Tableau, build a dashboard combining a bar chart showing department distribution and a scatter plot displaying the relationship between years of service and performance scores.

4. In R, develop a table showing the performance data for each employee. Label the table elements.


RGui
File Edit Packages Windows Help
R Console
C:\Data Handling\Lab5.R - R Editor
22.
1.
Month <- c("Jan", "Feb", "Mar", "Apr", "May")
Sales <- c(15000, 18000, 22000, 20000, 23000)
sales_ts <- ts(Sales, start = 1, frequency = 12)
plot(sales_ts, type = "o", main = "Monthly Sales Trend", xlab = "Month", ylab =
2.
Budget <- c(1000, 1500, 2000, 1800, 2500)
plot(Budget, Sales, pch = 16, col = "blue", main = "Budget vs Sales", xlab = "
3.
acf(sales_ts, main = "Autocorrelation of Sales")
23.
1.
Dept <- c("Sales", "HR", "Marketing", "Sales", "HR")
counts <- table(Dept)
barplot(counts, main = "Employee Count by Department", xlab = "Department", yla
R Graphics: Device 2 (ACTIVE)
Employee Count by Department
Number of Employees
0.0
0.5
1.0
1.5
2.0
HR
Marketing
Sales
Department
35°C
Partly sunny
Search
ENG
US
49%
13:44
12-08-2026


RGui
File Edit Packages Windows Help
R Console
> sales_ts <- ts(Sales, start = 1, frequency = 12)
> plot(sales_ts, type = "o", main = "Monthly Sales Trend", xlab = "Month", ylab$
> Budget <- c(1000, 1500, 2000, 1800, 2500)
> plot(Budget, Sales, pch = 16, col = "blue", main = "Budget vs Sales", xlab = $
> acf(sales_ts, main = "Autocorrelation of Sales")
> Dept <- c("Sales", "HR", "Marketing", "Sales", "HR")
> counts <- table(Dept)
> barplot(counts, main = "Employee Count by Department", xlab = "Department", y$
> Years <- c(5, 3, 7, 4, 2)
> Score <- c(85, 92, 78, 90, 76)
> order_idx <- order(Years)
> plot(Years[order_idx], Score[order_idx], type = "o", main = "Performance vs Y$
> Emp_ID <- c(1, 2, 3, 4, 5)
> Dept <- c("Sales", "HR", "Marketing", "Sales", "HR")
> Years <- c(5, 3, 7, 4, 2)
> Score <- c(85, 92, 78, 90, 76)
> performance_table <- data.frame(ID = Emp_ID, Department = Dept, Service_Years$
> print(performance_table)
ID Department Service_Years Performance_Score
1 1 Sales 5 85
2 2 HR 3 92
3 3 Marketing 7 78
4 4 Sales 4 90
5 5 HR 2 76
C:\Data Handling\Lab5.R - R Editor
acf(sales_ts, main = "Autocorrelation of Sales")
23.
1.
Dept <- c("Sales", "HR", "Marketing", "Sales", "HR")
counts <- table(Dept)
barplot(counts, main = "Employee Count by Department", xlab = "Department", yla
2.
Years <- c(5, 3, 7, 4, 2)
Score <- c(85, 92, 78, 90, 76)
order_idx <- order(Years)
plot(Years[order_idx], Score[order_idx], type = "o", main = "Performance vs Yea
4.
Emp_ID <- c(1, 2, 3, 4, 5)
Dept <- c("Sales", "HR", "Marketing", "Sales", "HR")
Years <- c(5, 3, 7, 4, 2)
Score <- c(85, 92, 78, 90, 76)
performance_table <- data.frame(ID = Emp_ID, Department = Dept, Service_Years =
print(performance_table)
35°C
Partly sunny
Search
ENG
US
48%
13:45
12-08-2026



Set 24 Product Inventory Management

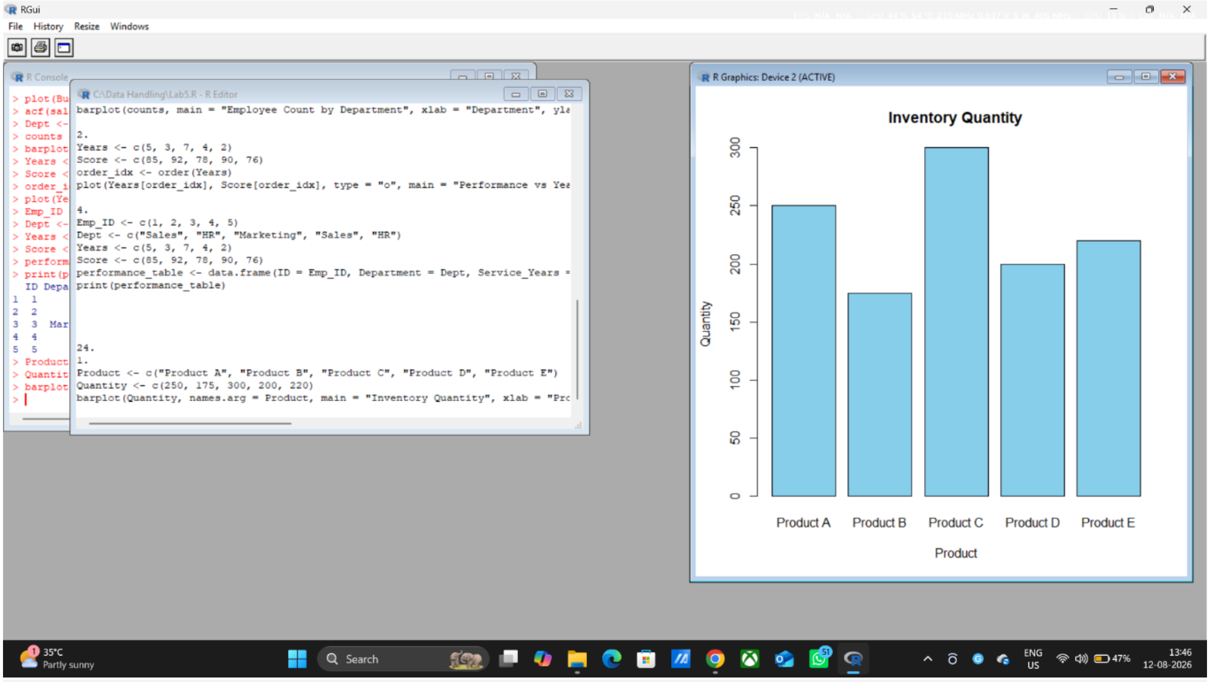
Dataset: Product Inventory

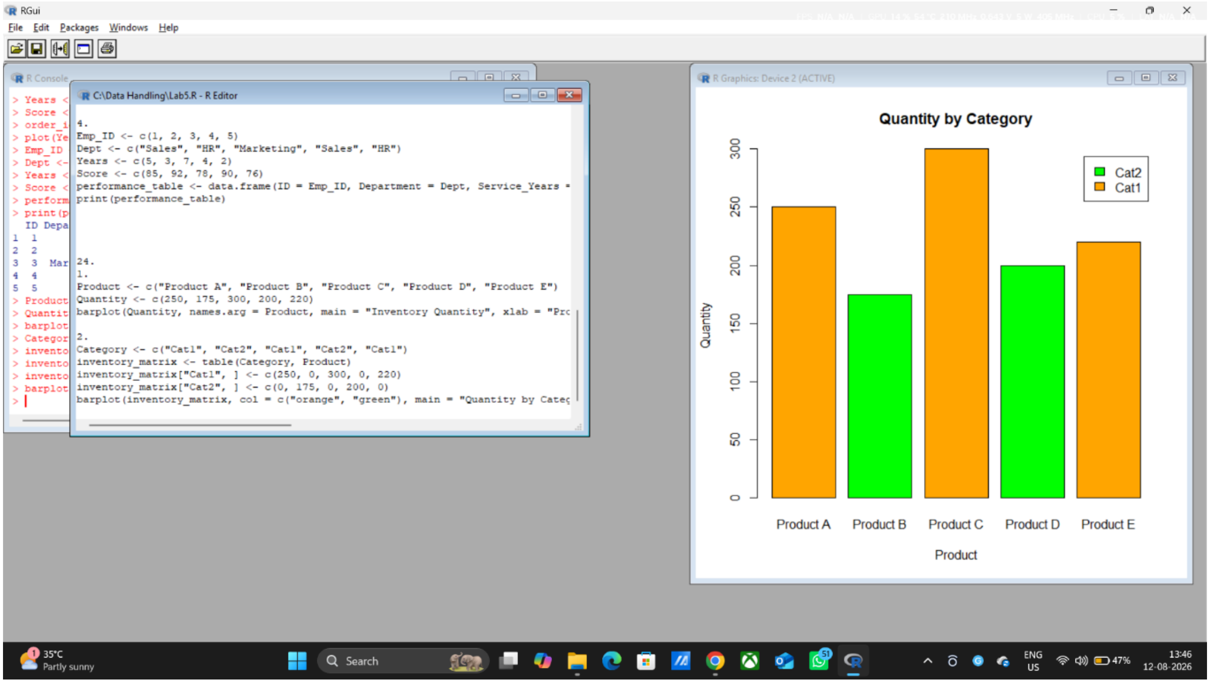
1.In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

2.In R, generate a stacked bar chart to show the quantity of each product within different product categories.

3.In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive exploration of inventory data.

4. In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights


RGui
File History Resize Windows
R Console
C:\Data Handling\Lab5.R - R Editor
barplot(counts, main = "Employee Count by Department", xlab = "Department", yla
2.
Years <- c(5, 3, 7, 4, 2)
Score <- c(85, 92, 78, 90, 76)
order_idx <- order(Years)
plot(Years[order_idx], Score[order_idx], type = "o", main = "Performance vs Yea
4.
Emp_ID <- c(1, 2, 3, 4, 5)
Dept <- c("Sales", "HR", "Marketing", "Sales", "HR")
Years <- c(5, 3, 7, 4, 2)
Score <- c(85, 92, 78, 90, 76)
performance_table <- data.frame(ID = Emp_ID, Department = Dept, Service_Years =
print(performance_table)
24.
1.
Product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
Quantity <- c(250, 175, 300, 200, 220)
barplot(Quantity, names.arg = Product, main = "Inventory Quantity", xlab = "Pro
R Graphics: Device 2 (ACTIVE)
Inventory Quantity
Quantity
0
50
100
150
200
250
300
Product A
Product B
Product C
Product D
Product E
Product
35°C
Partly sunny
Search
ENG
US
47%
13:46
12-08-2026


RGui
File Edit Packages Windows Help
R Console
C:\Data Handling\Lab5.R - R Editor
4.
Emp_ID <- c(1, 2, 3, 4, 5)
Dept <- c("Sales", "HR", "Marketing", "Sales", "HR")
Years <- c(5, 3, 7, 4, 2)
Score <- c(85, 92, 78, 90, 76)
performance_table <- data.frame(ID = Emp_ID, Department = Dept, Service_Years =
print(performance_table)
24.
1.
Product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
Quantity <- c(250, 175, 300, 200, 220)
barplot(Quantity, names.arg = Product, main = "Inventory Quantity", xlab = "Pro
2.
Category <- c("Cat1", "Cat2", "Cat1", "Cat2", "Cat1")
inventory_matrix <- table(Category, Product)
inventory_matrix["Cat1",] <- c(250, 0, 300, 0, 220)
inventory_matrix["Cat2",] <- c(0, 175, 0, 200, 0)
barplot(inventory_matrix, col = c("orange", "green"), main = "Quantity by Categ
R Graphics: Device 2 (ACTIVE)
Quantity by Category
Cat2
Cat1
Quantity
0
50
100
150
200
250
300
Product A
Product B
Product C
Product D
Product E
Product
35°C
Partly sunny
Search
ENG
US
47%
13:46
12-08-2026

Set 25: Website Traffic Analysis

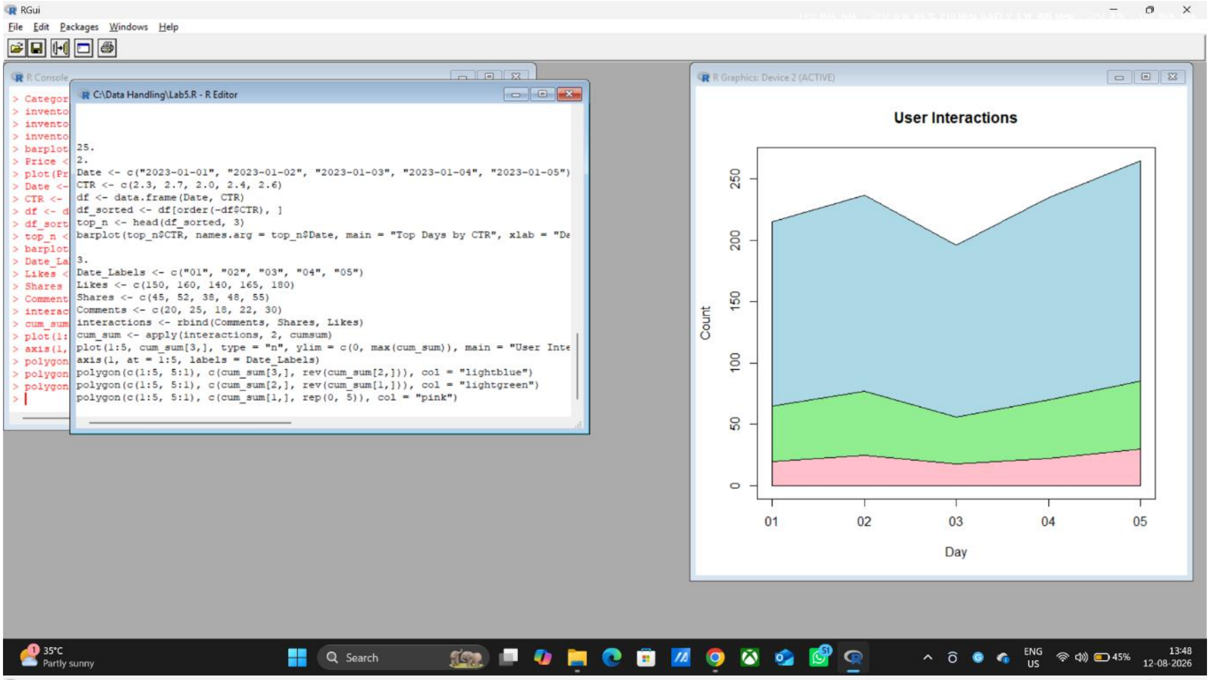
Dataset: Website Traffic

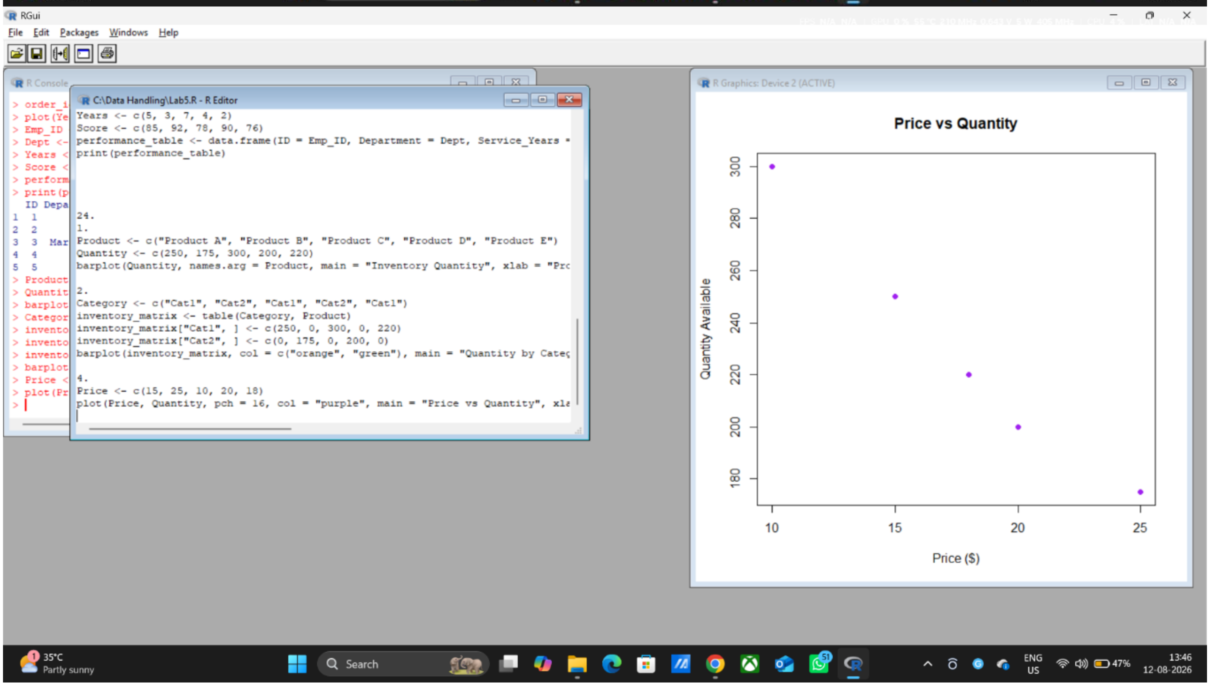
1. In Tableau, create a line chart to visualize the trend in page views over time. Label the axes and title the chart.

2. In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.

3. In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.

4. In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source..


RGui
File Edit Packages Windows Help
R Console
C:\Data Handling\Lab5.R - R Editor
25.
2.
Date <- c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")
CTR <- c(2.3, 2.7, 2.0, 2.4, 2.6)
df <- data.frame(Date, CTR)
df_sorted <- df[order(-df$CTR),]
top_n <- head(df_sorted, 3)
barplot(top_n$CTR, names.arg = top_n$Date, main = "Top Days by CTR", xlab = "Da
3.
Date_Labels <- c("01", "02", "03", "04", "05")
Likes <- c(150, 160, 140, 165, 180)
Shares <- c(45, 52, 38, 48, 55)
Comments <- c(20, 25, 18, 22, 30)
interactions <- rbind(Comments, Shares, Likes)
cum_sum <- apply(interactions, 2, cumsum)
plot(1:5, cum_sum[3,], type = "n", ylim = c(0, max(cum_sum)), main = "User Inte
axis(1, at = 1:5, labels = Date_Labels)
polygon(c(1:5, 5:1), c(cum_sum[3,], rev(cum_sum[2,])), col = "lightblue")
polygon(c(1:5, 5:1), c(cum_sum[2,], rev(cum_sum[1,])), col = "lightgreen")
polygon(c(1:5, 5:1), c(cum_sum[1,], rep(0, 5)), col = "pink")
R Graphics: Device 2 (ACTIVE)
User Interactions
Count
0
50
100
150
200
250
01
02
03
04
05
Day
35°C
Partly sunny
Search
ENG
US
45%
13:48
12-08-2026


RGui
File Edit Packages Windows Help
R Console
C:\Data Handling\Lab5.R - R Editor
Years <- c(5, 3, 7, 4, 2)
Score <- c(85, 92, 78, 90, 76)
performance_table <- data.frame(ID = Emp_ID, Department = Dept, Service_Years =
print(performance_table)
24.
1.
Product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
Quantity <- c(250, 175, 300, 200, 220)
barplot(Quantity, names.arg = Product, main = "Inventory Quantity", xlab = "Pro
2.
Category <- c("Cat1", "Cat2", "Cat1", "Cat2", "Cat1")
inventory_matrix <- table(Category, Product)
inventory_matrix["Cat1",] <- c(250, 0, 300, 0, 220)
inventory_matrix["Cat2",] <- c(0, 175, 0, 200, 0)
barplot(inventory_matrix, col = c("orange", "green"), main = "Quantity by Categ
4.
Price <- c(15, 25, 10, 20, 18)
plot(Price, Quantity, pch = 16, col = "purple", main = "Price vs Quantity", xla
R Graphics: Device 2 (ACTIVE)
Price vs Quantity
Quantity Available
180
200
220
240
260
280
300
10
15
20
25
Price ($)
35°C
Partly sunny
Search
ENG
US
47%
13:46
12-08-2026